Certyfikat PMP: Project Scope

Initiation, Selection, and the Genesis of Project Authorization

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> “The definition of a successful project is one where the stakeholders are satisfied.

PHASE 1: ACTIVE

STAKEHOLDER ALIGNMENT: IN PROGRESS

PROJECT INITIATION OVERVIEW

Initiation is the formal recognition that a project should begin. It is where needs, expectations, and scope boundaries are first recognized to ensure ultimate satisfaction.

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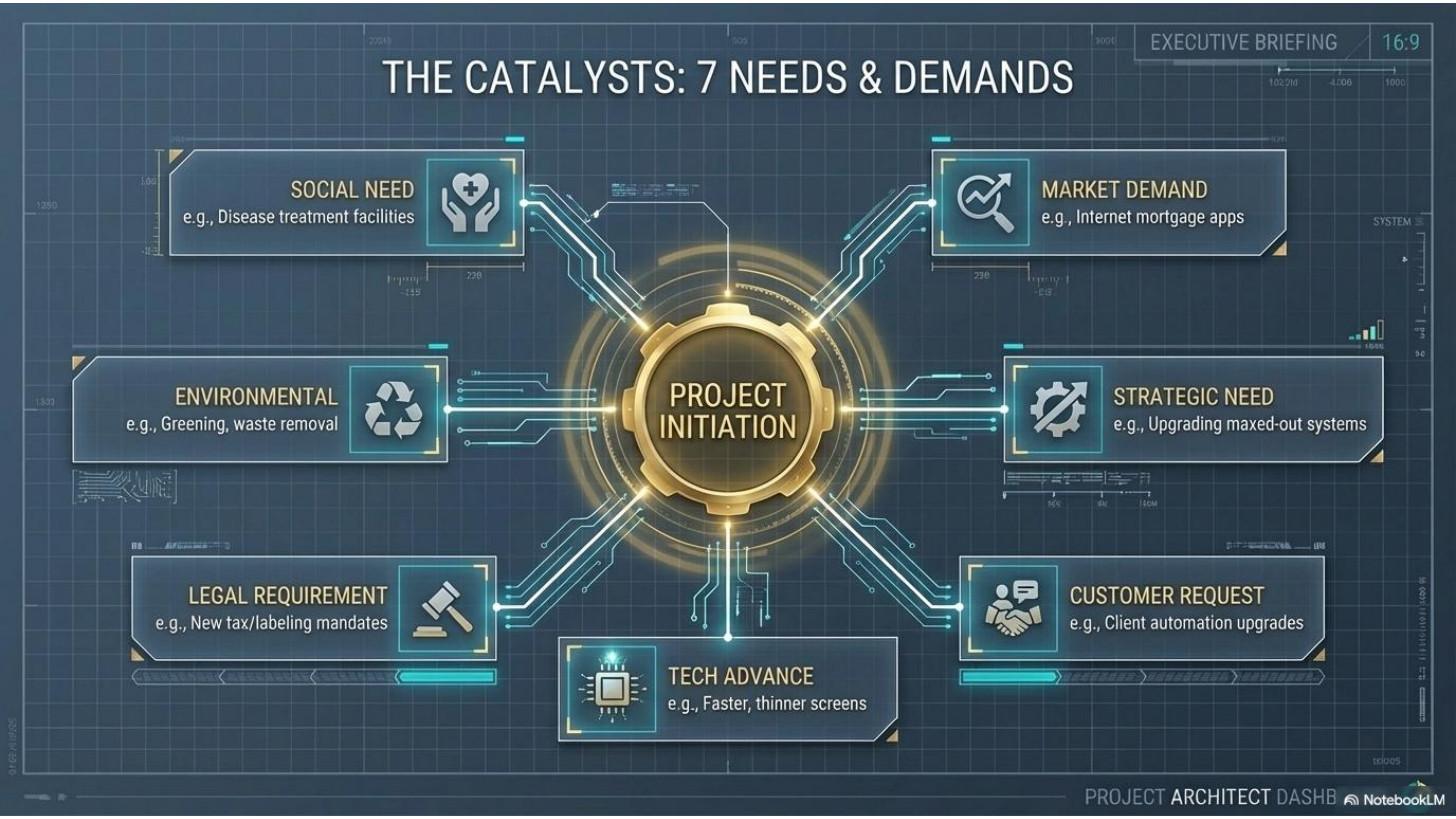
EXECUTIVE BRIEFING
16:9
THE CATALYSTS: 7 NEEDS & DEMANDS
SOCIAL NEED
e.g., Disease treatment facilities
MARKET DEMAND
e.g., Internet mortgage apps
ENVIRONMENTAL
e.g., Greening, waste removal
PROJECT INITIATION
STRATEGIC NEED
e.g., Upgrading maxed-out systems
LEGAL REQUIREMENT
e.g., New tax/labeling mandates
CUSTOMER REQUEST
e.g., Client automation upgrades
TECH ADVANCE
e.g., Faster, thinner screens
PROJECT ARCHITECT DASHB
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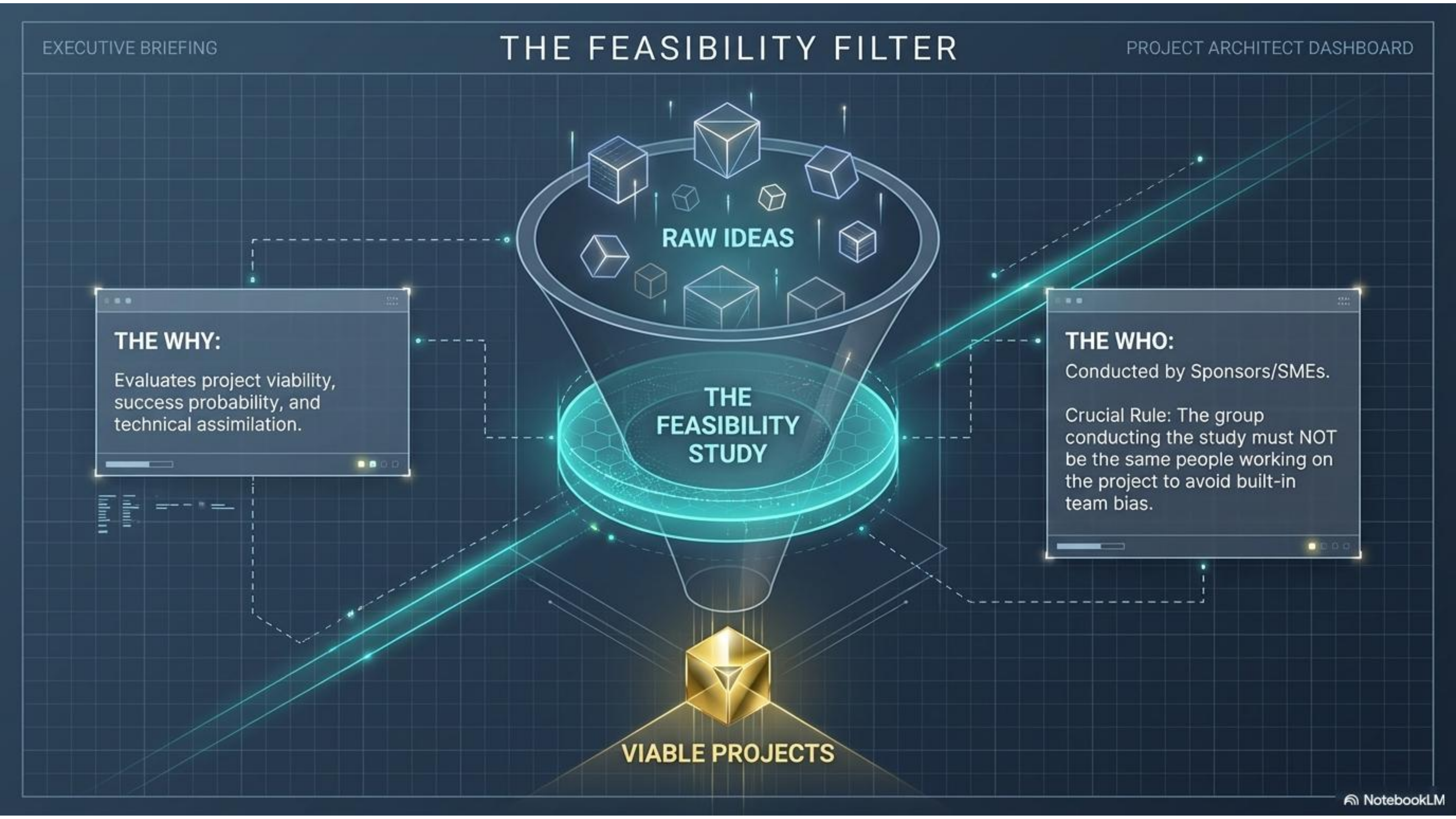
EXECUTIVE BRIEFING
THE FEASIBILITY FILTER
PROJECT ARCHITECT DASHBOARD
RAW IDEAS
THE WHY:
Evaluates project viability, success probability, and technical assimilation.
THE FEASIBILITY STUDY
THE WHO:
Conducted by Sponsors/SMEs.
Crucial Rule: The group conducting the study must NOT be the same people working on the project to avoid built-in team bias.
VIABLE PROJECTS
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THE PROJECT SELECTION DIVIDE

MATHEMATICAL MODELS

Constrained Optimization Methods

- **Mechanics**: Linear, dynamic, integer, nonlinear, and multi-objective programming.
- **Complexity**: Requires statistical/engineering backgrounds; relies on complex algorithms.

BENEFIT MEASUREMENT METHODS

Decision Models

PMP EXAM FOCUS

- **Mechanics**: Comparative approaches, economic models, scoring models.
- **Focus**: The vast majority of PMP project selection techniques fall into this category.

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BENEFIT MEASUREMENT: SCORING MODELS

SCORING DASHBOARD

Criteria	Weight	Project A Score	Project A Total	Project B Score	Project B Total
Profit potential	5	5	25	5	25
Marketability	3	4	12	3	9
Ease to produce	1	4	4	3	3
Weighted Score	–	–	41 (WINNER)	–	37

WEIGHT ASSIGNMENT

CALCULATION ENGINE

Weighted Scoring Models allow strategic importance to dictate the math. Heavier weights ensure alignment with corporate goals.

WEIGHT ASSIGNMENT

STRATEGIC ALIGNMENT

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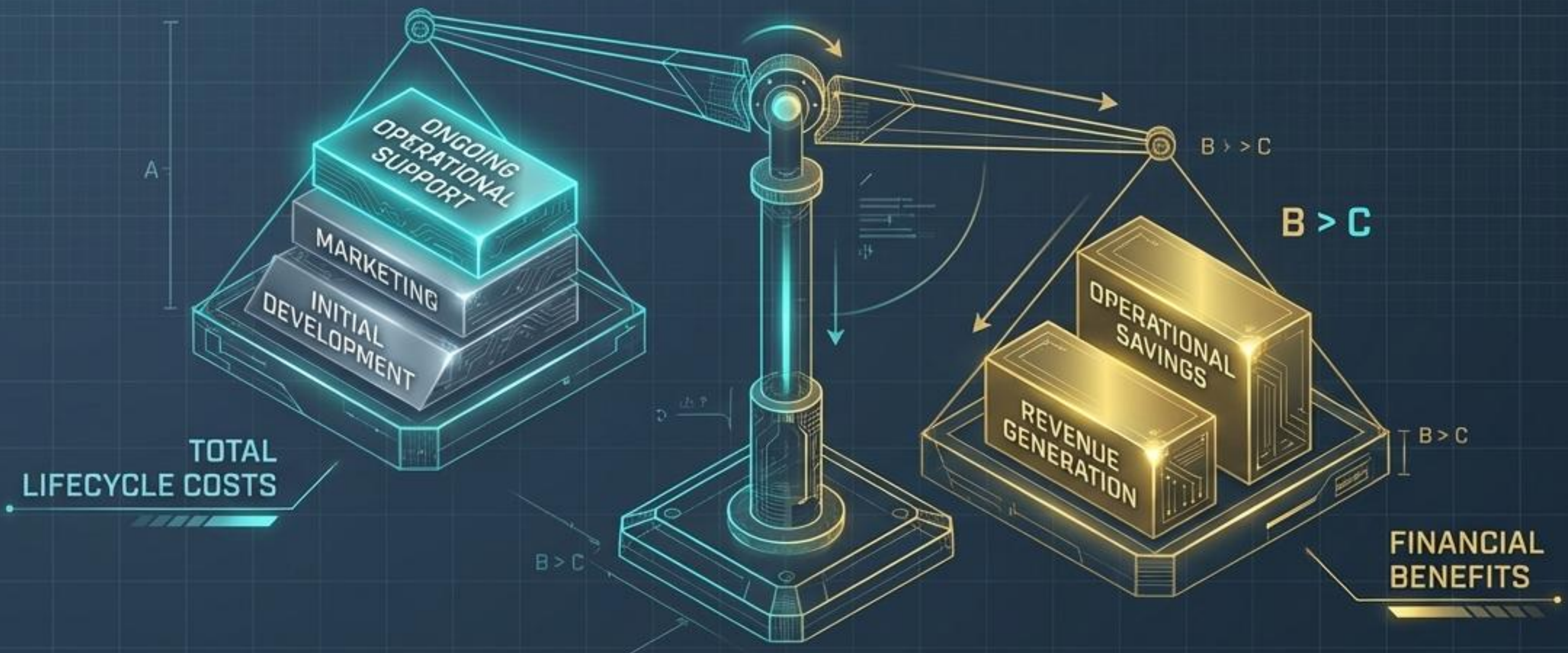
BENEFIT MEASUREMENT: COST-BENEFIT ANALYSIS (CBA)



STRATEGIC ALERT

The Golden Rule of CBA: Financial benefits must outweigh the total lifecycle costs. A common failure point is ignoring the ongoing operational support costs once the product goes to market.

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THE CORE CONCEPT: TIME VALUE OF MONEY

Money received in the future is worth less than money received today due to lost investment potential.

Future Value (FV)	Present Value (PV)
$FV = PV(1+i)^n$	$PV = \dfrac{FV}{(1+i)^n}$

Example: \$2,315 received in three years is only worth \$2,000 today (at a 5% interest rate).

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Cash Flow Evaluation Matrix

Technique	Definition	Precision Level	Time Value Considered?	Selection Rule
Payback Period	Time to recoup initial investment	Lowest	No	Shorter is better
Discounted Cash Flows (DCF)	Compares future cash flows to today's dollars	Medium	Yes	Highest total wins
Net Present Value (NPV)	Evaluates cash inflows using DCF, minus initial investment	High (Conservative)	Yes	Accept if > 0
Internal Rate of Return (IRR)	The discount rate when NPV equals 0	High (Complex)	Yes	Highest percentage wins

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EXECUTIVE DECISION DASHBOARD (NPV & IRR)

NET PRESENT VALUE (NPV)

< 0

> 0

- Rule: NPV > 0 = ACCEPT.
  NPV < 0 = REJECT.
- Nuance: Projects with high returns early in the lifecycle are superior.

INTERNAL RATE OF RETURN (IRR)

PROJECT A:	PROJECT B:
8%	3%

- Rule: IRR is the discount rate when NPV equals 0.
- Nuance: Always choose the project with the highest IRR.

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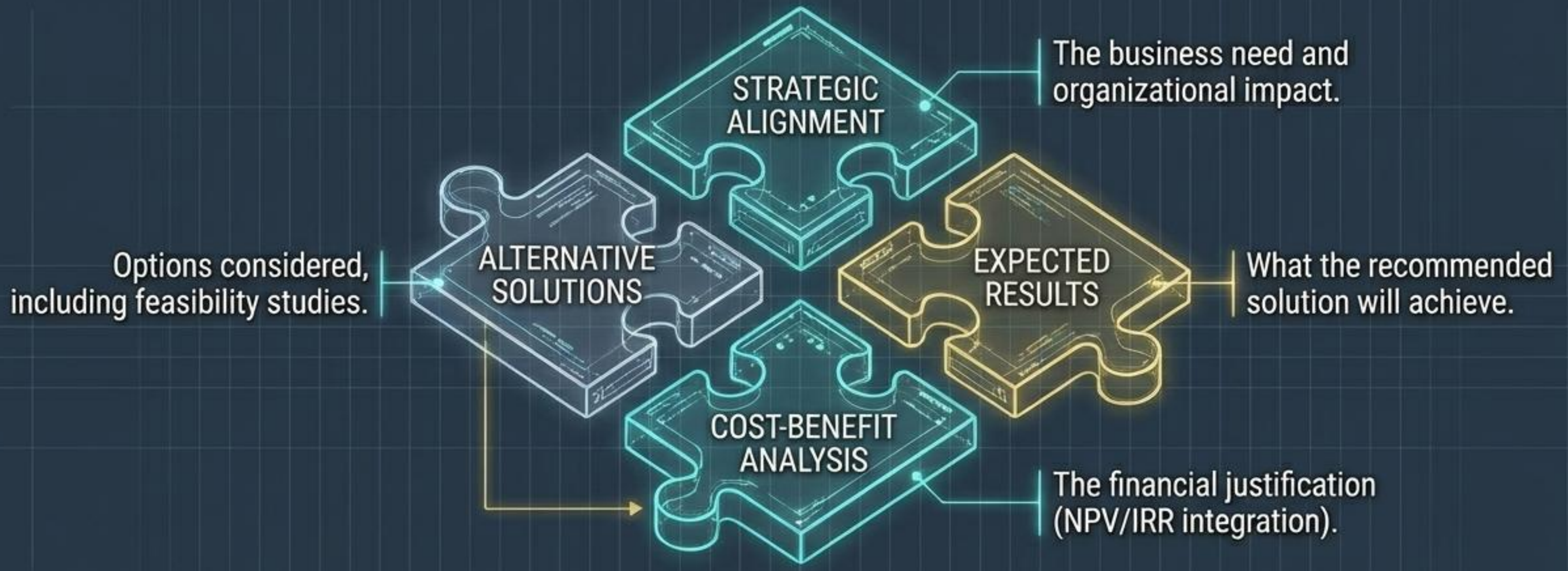
BUILDING THE BUSINESS CASE



BOUNDARY WARNING: The Business Case analysis, funding discussions, and approval all happen OUTSIDE the boundaries of the actual project. The project team is typically not involved yet.

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THE STATEMENT OF WORK (SOW)

PROGRESSIVE ELABORATION:
Scope is progressively elaborated. Product descriptions contain less detail in the early phases and gain maximum detail during Executing.

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THE PROJECT CHARTER

CORE FUNCTION OF THE CHARTER:

- It is the written acknowledgment that the project officially exists.
- The Critical Authority Rule: The Charter officially names the Project Manager and grants them the organizational authority to commit resources. The project does not begin until this is signed.

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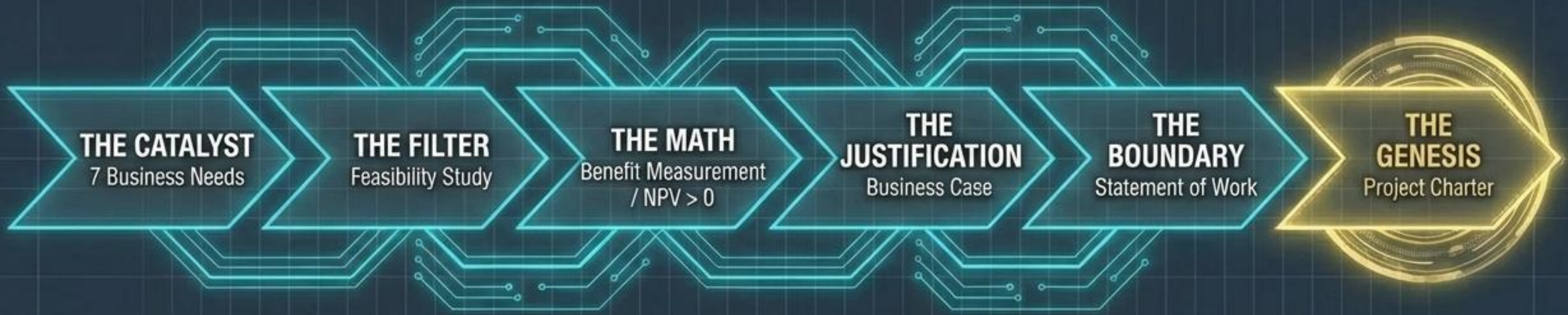
SYNTHESIS: THE SCOPE GENESIS



Project Scope is never created in a vacuum. It is progressively elaborated and rigorously justified before the first resource is ever assigned.

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FINAL TAKEAWAYS

THE SELECTION BOUNDARY

Formal project selection is technically outside the Project Manager's scope (handled by sponsors/committees), but PMs must understand the math to assist.

CASH IS KING

The Time Value of Money dictates decision-making. Always favor projects with NPV > 0 and the highest Internal Rate of Return (IRR).

THE STARTING LINE

The Project Charter is the ultimate output of Initiation. It bridges business strategy strategy with project execution and officially empowers the PM.

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